

Batman Attempts

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1 Early Stopping

First of all, I did the early stopping. To quickly recap, my constraints are listed as follows:

```
d_new = jnp.maximum(d_new, 0.05)
c_new = jnp.maximum(c_new, 0.5)
a_new = jnp.clip(a_new, 1.0, 3.0)
# Restrict b instead of a
max_b = d_new * (2.0 * a_new - 1.0)
b_new = jnp.clip(b_new, 0.0, max_b)
```

Figure 1: Constraint set 3.

I decided to keep a loose constraint on a so I can let it fluctuate more freely and pick the optimal parameter with lowest variance. Here are the results:

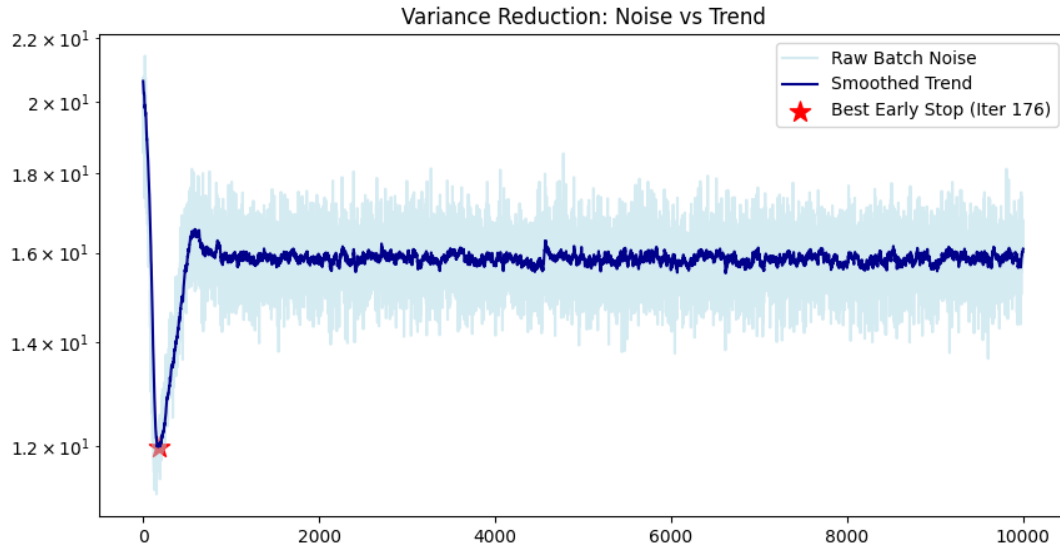


Figure 2: The variance change starting from $\theta = [2.0, 0.5, 1.0, 0.2]$.

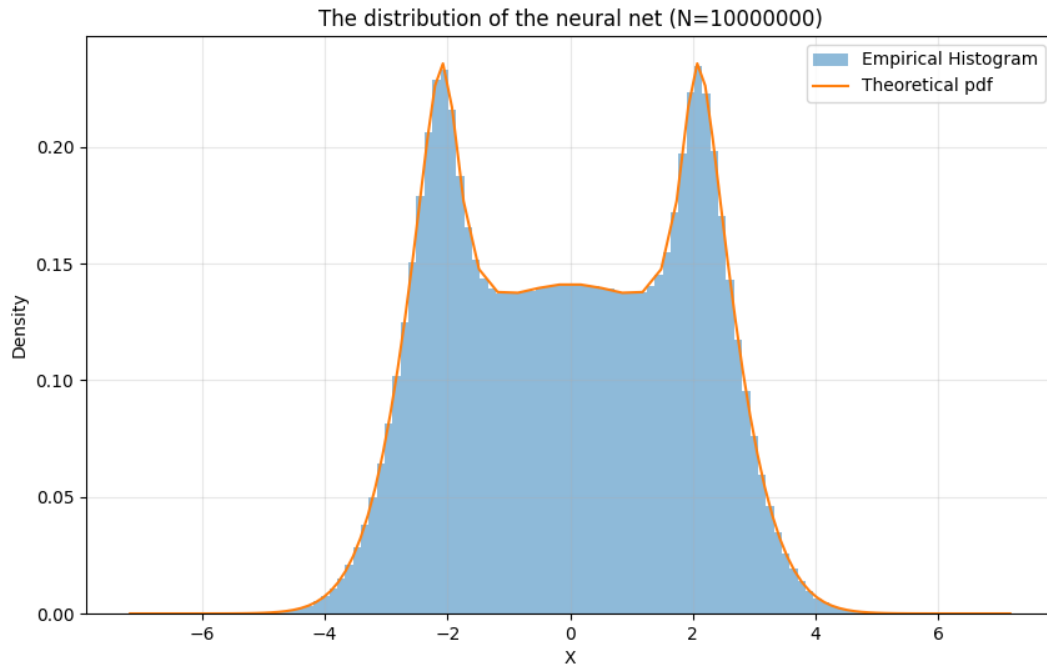


Figure 3: The distribution of the neural net after early stopping.

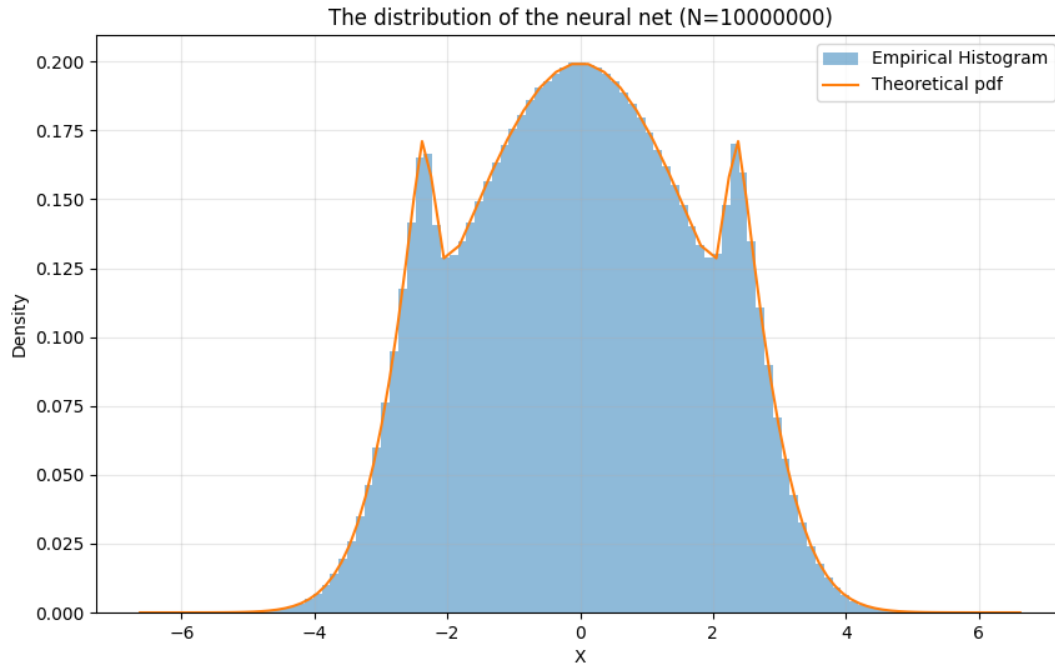


Figure 4: The distribution of the neural net without early stopping.

In deed, the early stopping gives a much better graph at $\theta = [1.42, 0.18, 0.64, 0.1]$. Although I did guessed a smaller constraint for a between 1.5 and 3.0, it's much better to let the optimizer do the work.

2 Mixing

This gives me a very bad and complicated score function. I'll show you the handwritten details when we meet.